Position priors dominate instance-level localisation metrics for weakly-supervised MIL in volumetric CT

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***Abstract*—Attention-based multiple-instance learning (MIL) is routinely credited with localising findings in volumetric CT on the strength of instance-level AUC against expert masks. We report a pre-registered ten-hypothesis evaluation of what that number measures. On LIDC-IDRI, flat 3D instance AUC is within-slice AUC in disguise: no scored arm separates the two by more than 0.02 except the one that injects an explicit axial prior, and a content-free reference fit to lesion masks alone reaches 0.8543, above every trained arm. Explicit prior injection is the single intervention that moves the measurement ($+0.1509$ patient-specific skill), and our instrument adjudicates that fix rather than competing with it. Restricted to a lung segmentation the position prior vanishes: the pre-registered template scores 0.4884 inside the lung. Which axis carries the prior is a property of the dataset — on MosMed the axial axis dominates (0.8455) exactly where LIDC's is empty. We recommend a one-line convention any paper can adopt: report raw and in-organ instance AUC side by side, and read the gap as that dataset's positional confound.**

***Index Terms*—multiple-instance learning, weakly-supervised localisation, computed tomography, evaluation methodology, pre-registration**

I. Introduction

Weakly-supervised MIL is trained on a bag-level label and asked, at evaluation time, to say *where* the finding is. The standard evidence is an instance-level AUC: score every instance by its attention weight, label instances by intersection with an expert mask, and pool. On volumetric CT the instances are patch tokens over a stack of slices, and the number is reported as three-dimensional localisation. That attention may do less work than assumed is already visible at the classification level, where mean pooling matches attention MIL across seven 3D tasks [1]; we ask the corresponding question of the localisation number, and answer with three claims.

(i) The reported 3D metric is largely in-plane anatomical position. Flat instance AUC over a whole volume is statistically indistinguishable from within-slice AUC for seven of eight scored arms. A content-free reference that reads no image at all — a 256-number in-plane template fit to validation *lesion masks*, the mask-fitted reference — scores above every trained arm, and an untrained network reaches 0.7666 at the 95th percentile.

(ii) The one intervention that adds real localisation is explicit prior injection, and two tests that were not designed to agree both find it. Harvey-style Normal Guidance [2] adds a KL term pulling the attention's slice marginal towards a moment-matched Normal, and is both the only arm whose patient-specific skill passes its falsifier and the only arm the axis gate flags. The critique fails for that arm; our instrument validates their fix against a stronger reference than the one they used.

(iii) Which axis carries the prior is a property of the dataset. We pre-registered a stereotypy prediction and the data reversed it with power to spare: COVID ground-glass is *more* positionally stereotyped than a lung nodule, on the axis LIDC leaves empty (Fig. 2).

II. Setup and Estimands

Data. LIDC-IDRI [3]: 999 series over 991 patients, split $700/149/150$ by patient at seed 2027, a split the protocol was never developed against. Frozen DINOv2 ViT-B/14 features [4], $16{\times}16$ patches per slice; instance labels from four-radiologist consensus masks rasterised to the grid. MosMed [5] enters for one hypothesis only (50 of 1110 studies carry masks). Four conditions are declared; `nodule_present` carries the hypothesis family, and "all three conditions" below means the three LIDC conditions (companion, §3).

Arms. Ten: mean pooling, ABMIL [6], gated ABMIL, multi-head ABMIL, CLAM-SB [7], DSMIL [8], TransMIL [9], slot attention [10] with a diversity term, a fixed centre Gaussian, and Normal Guidance [2]; five seeds each; eight are scored. Every AUC is a patient-clustered bootstrap over per-bag AUCs, 10,000 replicates.

Two templates, two names. Two objects here are each an in-plane template fit on validation. The *mask-fitted reference* is fit to *lesion masks*: it reads no image or attention and sets the content-free ceiling (0.8302 in-plane, 0.8543 separable). The *attention-fitted denominator* is the same rule fit to *the arm's own attention*: it is the pre-registered reference A_t of prior-normalised skill $= (A - A_t)/(1 - A_t)$, and its median over the scored dumps is 0.4208. The first is an oracle no arm reaches; the second adapts to the arm being scored. Patient-specific skill is $A_{\text{real}} - A_{\text{cross-patient}}$: the score that survives swapping whose masks the attention is scored against [11, 12].

III. Results

A. The floor, the ceiling, and the axis (H1–H3)

$|\text{flat} - \text{within-slice}|$ exceeds 0.02 for one of eight arms (the falsifier required a majority); the positive controls separate (`masks:axial` 0.1064, centre prior 0.0518), so the test can detect axial content when present. The one arm over the

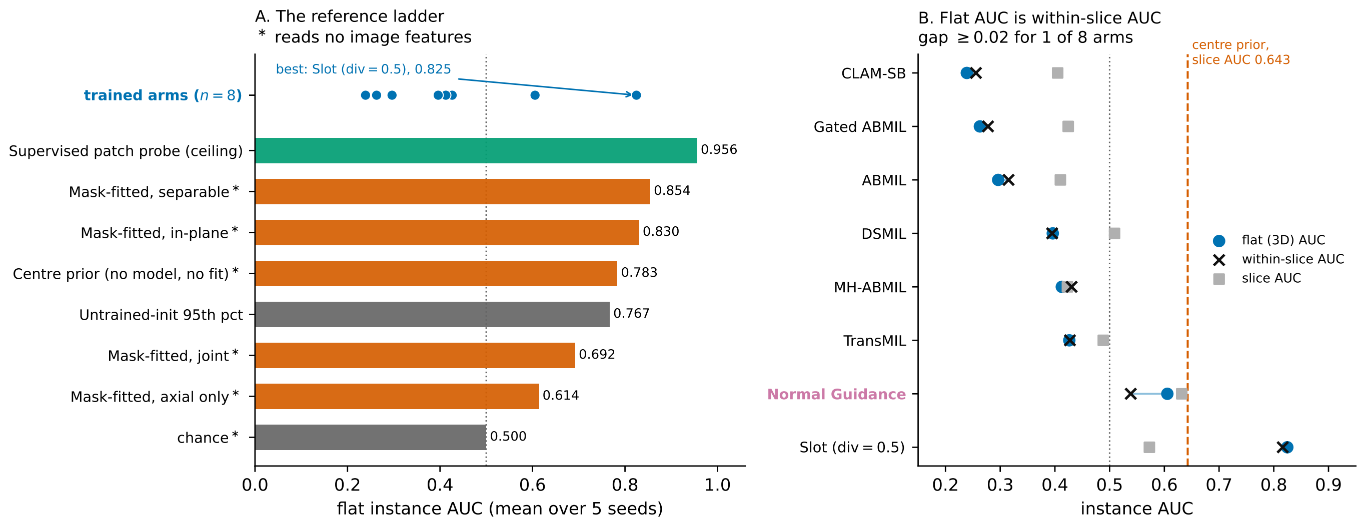

Fig. 1. **Left:** the reference ladder on `nodule_present`. Starred references are mask-fitted references: they read no image features and are computed from lesion masks and patch geometry. The best trained arm — the slot arm, 0.8247 — sits below the separable mask-fitted reference and barely above an untrained network's 95th percentile. **Right:** flat, within-slice and slice AUC per arm; flat and within-slice coincide for every arm but Normal Guidance, and no arm's slice AUC exceeds the model-free centre prior.

bar is Normal Guidance (0.0669, next-highest 0.0190). No trained arm reaches the centre prior's slice AUC of 0.6427; the best, Normal Guidance at 0.6318, is a statistical tie (paired per-bag Δ, patient-clustered over five seeds: $+0.0109$ $[-0.0078, +0.0300]$), and every other arm's interval sits entirely on the prior's side. The 95th-percentile instance AUC over 30 untrained initialisations is 0.7666 against a bar of 0.70: chance is not the floor, and a reported 0.79–0.82 is inside or barely above the untrained range.

B. What survives normalisation (H4, H5)

Median patient-specific skill is -0.0466 and five of eight arms are negative: scoring an arm's attention against *another patient's* masks beats scoring it against the right ones. No arm's prior-normalised skill exceeds 0.30 (max $+0.1990$); one seed sits at 0.3432 and is reported rather than absorbed, because the unit was declared before that seed existed. Because the declared attention-fitted denominator sits below chance for 24 of 40 dumps — which makes a stay-below-the-bar hypothesis easier to pass — the companion recomputes the identical statistic with the denominator floored at 0.5: no verdict moves, and flooring lowers the maximum wherever it binds.

C. Prior injection is the exception (H6)

The falsifier stated that if patient-specific skill rises by ≥ 0.02 under prior injection, the critique fails for that arm. It rose $+0.1509$, from -0.1019 to $+0.0490$; we report the failure. Three things make it the second finding. Magnitude: Normal Guidance is one of only three arms with positive patient-specific skill at all, and the only one that also clears the axis gate. Independence: the axis gate and the skill are different statistics on different artefacts, and both single out the same arm of eight. Mechanism: the arm's only departure from its base is a KL term naming the axial coordinate, so the gain cannot be attributed to a stronger backbone. Nothing in the architecture sweep adds localisation; a loss term that names the axis explicitly does — yet its slice AUC still only ties the centre prior, so prior injection moves the measurement without solving the problem.

D. The power gate, and its scope (H7)

The constructive half is gated: a supervised patch probe must clear 0.50 prior-normalised skill while every content-free baseline stays below 0.05. The gate passes on `nodule_present` (0.7305; content-free members ≤ 0.0105) and `malignancy` (0.7036), and fails on `balanced_presence` (0.4772): **we recommend the estimands only where the per-condition gate passes.** The gate's stochastic members were originally computed from a single shared realisation, under which H7 *failed*; the amendment replicating them (30 draws, mean over draws, the stricter aggregation left in force) was committed before the replicated numbers existed, and the full measurement history is in the companion.

E. The portable convention (H8)

H8 was declared two-sided, with it written in advance that if restricting to lung removed the prior, that becomes the recommendation rather than a defeat. It does: the lung is 25.2% of a bag, and inside it the pre-registered attention-fitted denominator scores 0.4884 against a bar of 0.65 (set against the unrestricted template's 0.786 on discovery), all three LIDC conditions agreeing. No arm's attention-fitted denominator beats chance inside the lung. Applied to the arms' own attention — Recommendation 1 on ourselves — the same restriction confirms the reading (companion): the strongest trained arm, the slot arm at flat 0.8247, falls to 0.5560 inside the lung — far the largest loss of the eight — and is also the arm whose template loses the most (-0.300) and the owner of H5's one

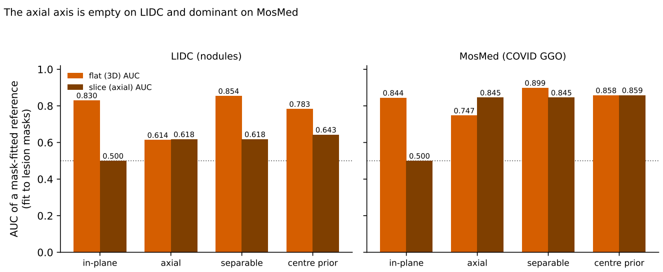

Fig. 2. Mask-fitted references, LIDC versus MosMed. The axial axis is empty on LIDC and dominant on MosMed; no training is involved in any bar.

over-bar seed. The "interpretable" arm looked best precisely because it learned lung geometry.

F. The dataset contrast (H9)

H9 predicted prior dominance would track positional stereotypy with the lung nodule as the more stereotyped finding; it was reversed with power to spare (median observed separation -0.2495 against a required $+0.0560$). COVID ground-glass is peripheral in-plane and basal axially, and MosMed's mask-fitted axial template reaches slice AUC 0.8455 against LIDC's 0.6180. Which axis carries the prior must be measured on the cohort being reported, and Fig. 2 costs no training to produce.

IV. Recommendations

1) **Report raw and in-organ instance AUC side by side**; the gap is that dataset's positional confound. One segmentation, no retraining.
2) **Report the axis decomposition** (flat, within-slice, slice). If flat equals within-slice, the metric is not measuring the third dimension.
3) **Measure the floor**: a content-free positional reference and an untrained-initialisation percentile. Chance is not the floor.
4) **Normalise against the attention-fitted denominator and gate it** on the per-condition supervised-probe test; where the gate fails, fall back to 1.

V. Pre-registration

Every threshold, arm and estimand was frozen in a machine-readable pre-registration (hash `4fd6e801157ecef5`) before any confirmatory number existed; analysis code reads parameters only from that file and raises on anything undeclared. The full falsifier table, verdicts across conditions, the dated amendment chain, the multiplicity analysis, and the declared-versus-reported diff are in the companion and at https://github.com/BhaveshThapar/SlotMIL.

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